JURNAL PRAKTIKUM

Nama : Silfiana Dwi Nabiilah

Praktikum Minggu ke : 14

Hari / Tanggal : Jumat, 28 April 2023

Judul Praktikum : 7 Segment

- Alat & Komponen yang digunakan :
  1. 7 buah Resistor 330Ω
  2. 1 buah 7 Segment Common Anode
  3. 1 buah 7 Segment Common Cathode
  4. 4 buah Logicstate
  5. 7 buah Logicprobe
  6. 7 buah IC 7407
  7. 4 buah IC 7404
  8. 8 buah IC 7411
  9. 4 buah IC 7432
  10. 3 buah IC 7475
  11. 1 buah IC 7472
  12. 1 buah IC 7447
  13. 1 buah IC 7448

- Soal & Hasil Pengerjaan :

1. Rancanglah rangkaian decoder “biner to octal” menggunakan tampilan seven segment active low.

➢ Tahapan pengerjaan :

a) Membuat tabel kebenaran.

Input Biner			Output Oktal Tampilan 7 Segment						
B2	B1	B0	a	b	c	d	e	f	g
0	0	0	0	0	0	0	0	0	1
0	0	1	1	0	0	1	1	1	1
0	1	0	0	0	1	0	0	1	0
0	1	1	0	0	0	0	1	1	0
1	0	0	1	0	0	1	1	0	0
1	0	1	0	1	0	0	1	0	0
1	1	0	0	1	0	0	0	0	0
1	1	1	0	0	0	1	1	1	1

b) Membuat persamaan Minterm dari tabel kebenaran dan sederhanakan persamaan minterm dengan K-map

$a = \sum(m1, m4)$

B2 \ B1 B0	00	01	11	10
0	0	1	0	0
1	1	0	0	0

$Persamaan = \overline{B1}(B2 \oplus B0)$

$b = \sum(m5, m6)$

B2 \ B1 B0	00	01	11	10
0	0	0	0	0
1	0	1	0	1

$Persamaan = B2(B1 \oplus B0)$

$c = \sum(m2)$

B2 \ B1 B0	00	01	11	10
0	0	0	0	1
1	0	0	0	0

$Persamaan = \overline{B2}B1\overline{B0}$

$d = \sum(m1, m4, m7)$

B2 \ B1 B0	00	01	11	10
0	0	1	0	0
1	1	0	1	0

$Persamaan = \overline{B2B1}B0 + B2(\overline{B1B0} + B1B0$

$$e = \sum(m1, m3, m4, m5, m7)$$

B2 \ B1 B0	00	01	11	10
0	0	1	1	0
1	1	1	1	0

$$Persamaan = B0 + B2\overline{B1}$$

$$f = \sum(m1, m2, m3, m7)$$

B2 \ B1 B0	00	01	11	10
0	0	1	1	1
1	0	0	1	0

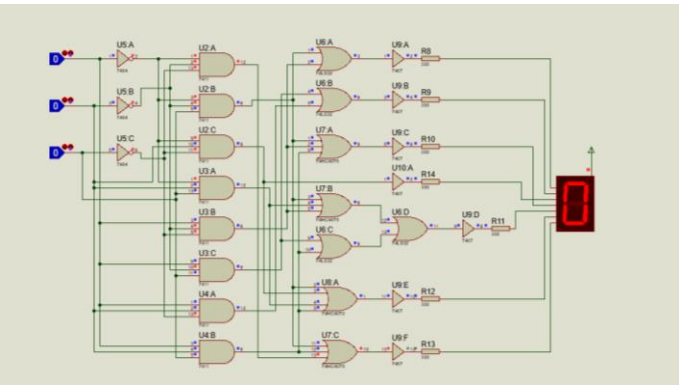
$$Persamaan = \overline{B2}(B0 + B1) + B1B0$$

$$g = \sum(m0, m1, m7)$$

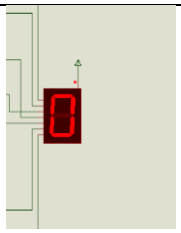
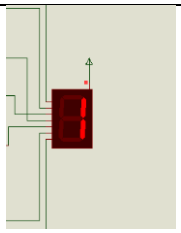
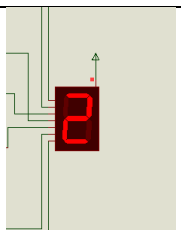
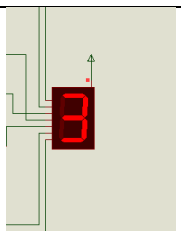
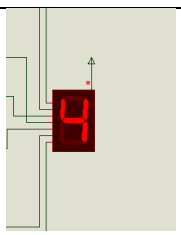
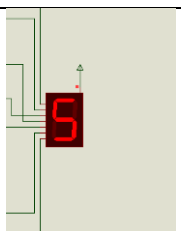
B2 \ B1 B0	00	01	11	10
0	1	1	0	0
1	0	0	1	0

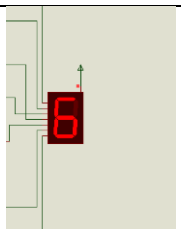
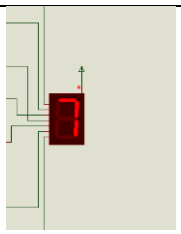
$$Persamaan = \overline{B2B1} + B2B1B0$$

- Screenshot gambar rangkaian :



➢ Tabel hasil :

Input Biner			Output Heksadesimal							Desimal
B2	B1	B0	a	b	c	d	e	f	g	
0	0	0	0	0	0	0	0	0	1	
0	0	1	1	0	0	1	1	1	1	
0	1	0	0	0	1	0	0	1	0	
0	1	1	0	0	0	0	1	1	0	
1	0	0	1	0	0	1	1	0	0	
1	0	1	0	1	0	0	1	0	0	

1	1	0	0	1	0	0	0	0	0	
1	1	1	0	0	0	1	1	1	1	

2. Rancanglah rangkaian decoder yang mempunyai input BCD dan 6 output sebagai berikut, menggunakan tampilan seven segment active high.

➢ Tahapan Pengerjaan :

a) Membuat tabel kebenaran

Input BCD				Output Heksadesimal Tampilan 7 Segment						
D	C	B	A	a	b	c	d	e	f	g
0	0	1	1	1	1	1	1	0	0	1
0	1	0	1	1	0	1	1	0	1	1
0	1	1	0	0	0	1	1	1	1	1
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	0	0	1	1
1	0	1	0	0	0	0	1	1	0	1
1	1	1	1	0	0	0	0	0	0	0

b) Membuat persamaan minterm dari tabel kebenaran dan sederhanakan persamaan minterm dengan K-map

$a = \sum(m3, m5, m8, m9)$

DC \ BA	00	01	11	10
00	0	0	1	0
01	0	1	0	0
11	0	0	0	0
10	1	1	0	0

$Persamaan = \overline{DC}BA + \overline{D}C\overline{B}A + D\overline{CB}$

$b = \sum(m3, m8, m9)$

DC \ BA	00	01	11	10
00	0	0	1	0
01	0	0	0	0
11	0	0	0	0
10	1	1	0	0

$Persamaan = \overline{DC}BA + D\overline{CB}$

$c = \sum(m3, m5, m6, m8, m9)$

DC \ BA	00	01	11	10
00	0	0	1	0
01	0	1	0	1
11	0	0	0	0
10	1	1	0	0

$Persamaan = \overline{DC}BA + \overline{D}C\overline{B}A + \overline{D}CB\overline{A} + D\overline{CB}$

$d = \sum(m3, m5, m6, m8, m9, m10)$

DC \ BA	00	01	11	10
00	0	0	1	0
01	0	1	0	1
11	0	0	0	0
10	1	1	0	1

$Persamaan = \overline{DC}BA + \overline{D}C\overline{B}A + \overline{D}CB\overline{A} + D\overline{CA}$

$e = \sum(m6, m8, m10)$

DC \ BA	00	01	11	10
00	0	0	0	0
01	0	0	0	1
11	0	0	0	0
10	1	0	0	1

$Persamaan = \overline{DC}BA + D\overline{CA}$

$f = \sum(m5, m6, m8, m9)$

DC \ BA	00	01	11	10
00	0	0	0	0
01	0	1	0	1
11	0	0	0	0
10	1	1	0	0

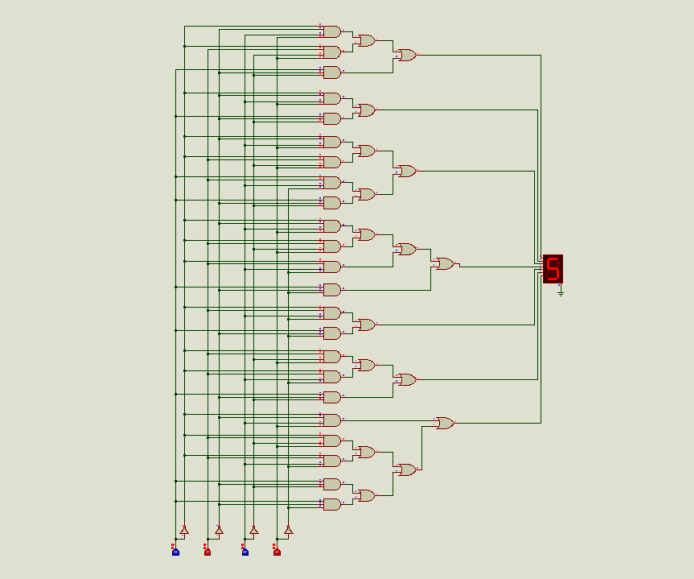
$Persamaan = \overline{DC}BA + \overline{D}CB\overline{A} + D\overline{CB}$

$g = \sum(m5, m6, m8, m9, m10)$

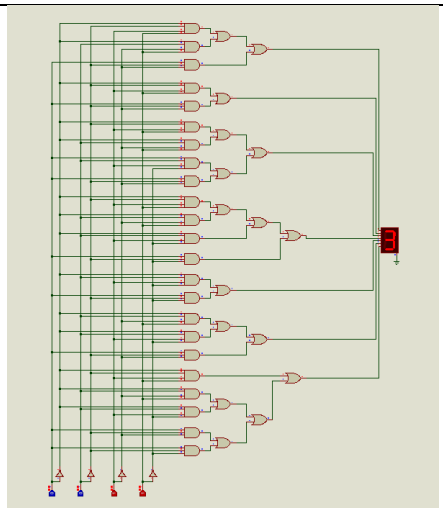
DC \ BA	00	01	11	10
00	0	0	0	0
01	0	1	0	1
11	0	0	0	0
10	1	1	0	1

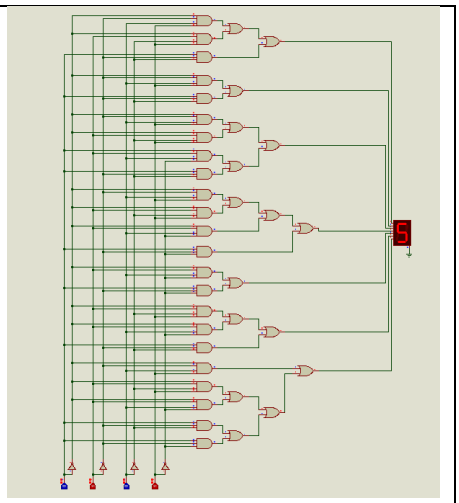
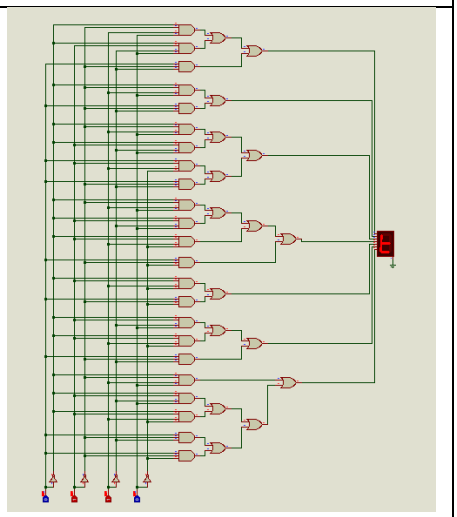
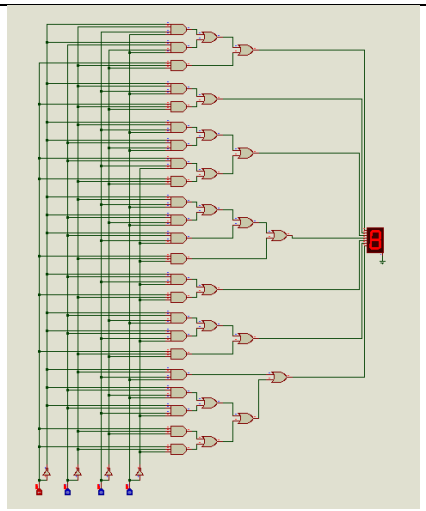
$Persamaan = \overline{DC}BA + \overline{D}C\overline{B}A + \overline{D}CB\overline{A} + D\overline{CB} + D\overline{CA}$

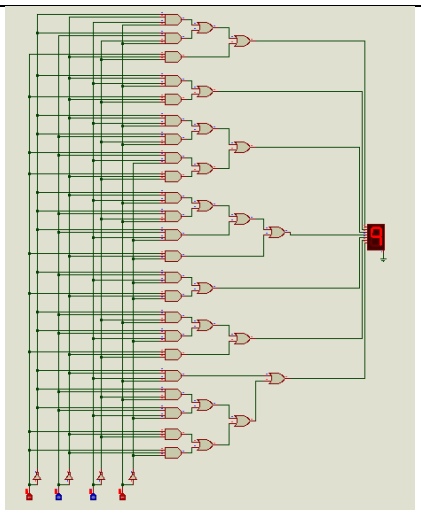
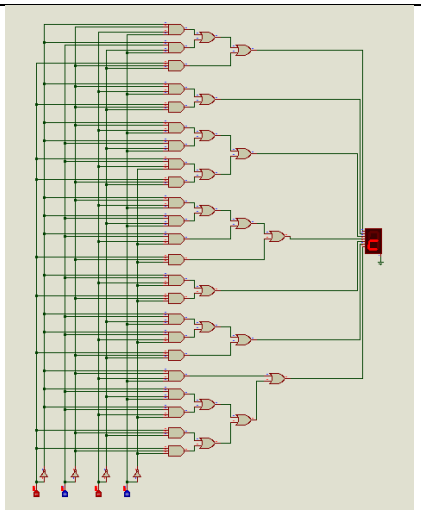
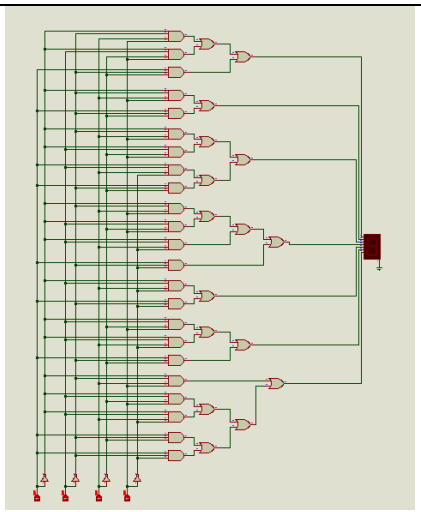
➢ Screenshot gambar rangkaian :



➢ Tabel hasil :

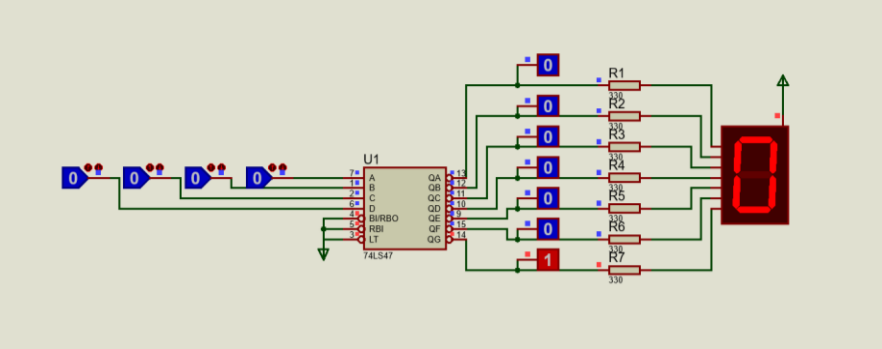
Input BCD				Output Heksadesimal Tampilan 7 Segment							Desimal
D	C	B	A	a	b	c	d	e	f	g	
0	0	1	1	1	1	1	1	0	0	1	

0	1	0	1	1	0	1	1	0	1	1	
0	1	1	0	0	0	1	1	1	1	1	
1	0	0	0	1	1	1	1	1	1	1	

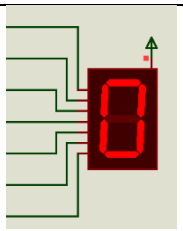
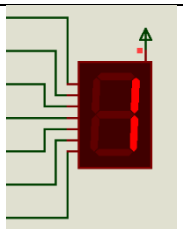
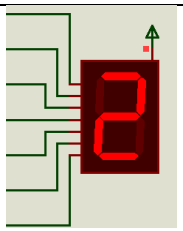
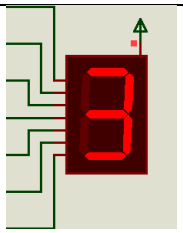
1	0	0	1	1	1	1	0	0	1	1	
1	0	1	0	0	0	0	1	1	0	1	
1	1	1	1	0	0	0	0	0	0	0	

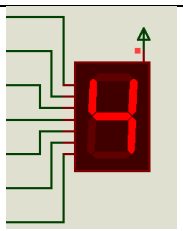
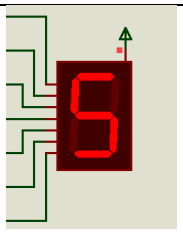
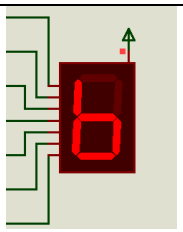
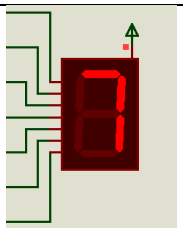
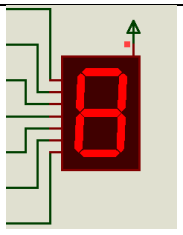
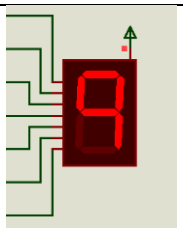
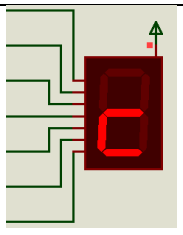
3. Buatlah rangkaian BCD to 7 Segment menggunakan IC 7447, dengan jenis 7 Segment common anode.

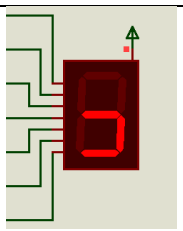
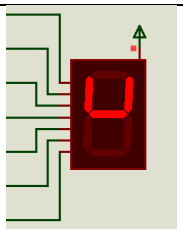
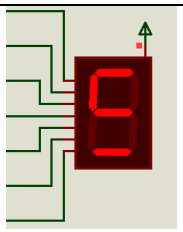
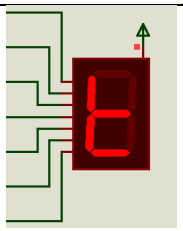
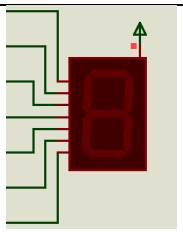
➢ Screenshot gambar rangkaian :



➢ Tabel hasil :

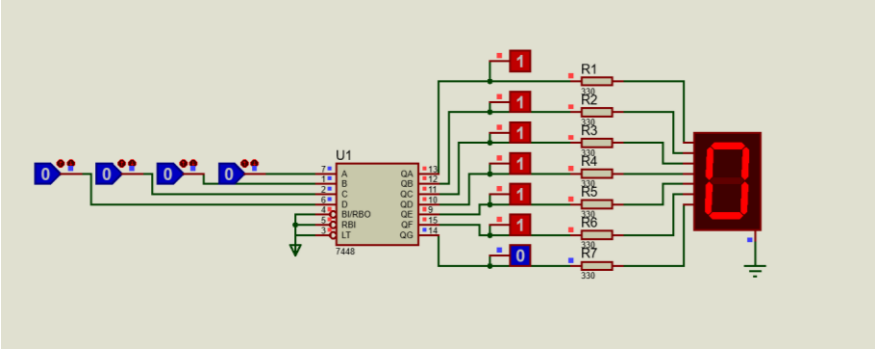
Input BCD				Output Heksadesimal							Desimal
D	C	B	A	a	b	c	d	e	f	g	
0	0	0	0	0	0	0	0	0	0	1	
0	0	0	1	1	0	0	1	1	1	1	
0	0	1	0	0	0	1	0	0	1	0	
0	0	1	1	0	0	0	0	1	1	0	

0	1	0	0	1	0	0	1	1	0	0	
0	1	0	1	0	1	0	0	1	0	0	
0	1	1	0	1	1	0	0	0	0	0	
0	1	1	1	0	0	0	1	1	1	1	
1	0	0	0	0	0	0	0	0	0	0	
1	0	0	1	0	0	0	1	1	1	0	
1	0	1	0	1	1	1	0	0	0	1	

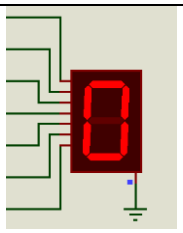
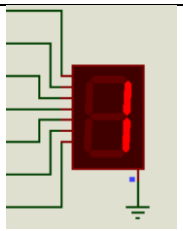
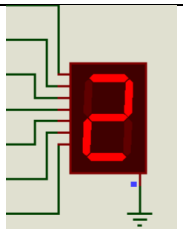
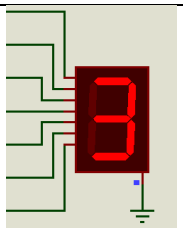
1	0	1	1	1	1	0	0	0	1	1	
1	1	0	0	1	0	1	1	1	1	0	
1	1	0	1	0	1	1	0	0	1	0	
1	1	1	0	1	1	1	0	0	0	0	
1	1	1	1	1	1	1	1	1	1	1	

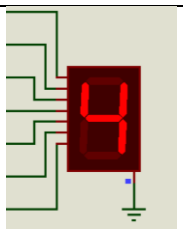
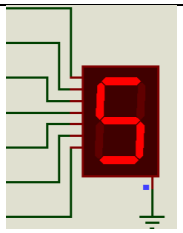
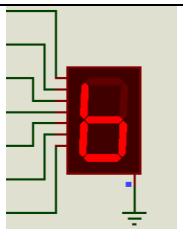
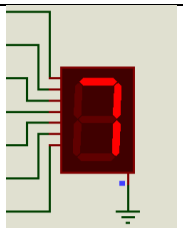
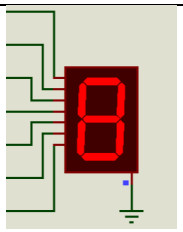
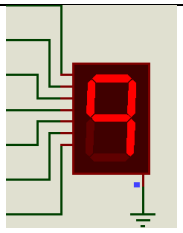
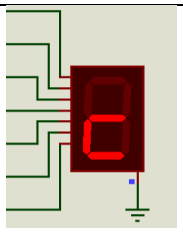
4. Buatlah rangkaian BCD to 7 Segment menggunakan IC 7448, dengan jenis 7 Segment common cathode.

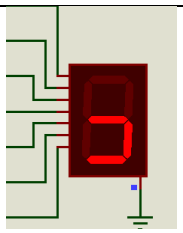
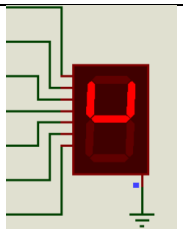
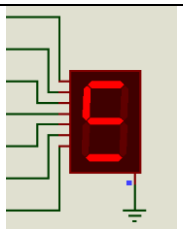
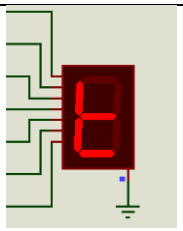
➢ Screenshot gambar rangkaian :



➢ Tabel hasil :

Input BCD				Output Heksadesimal							Desimal
D	C	B	A	a	b	c	d	e	f	g	
0	0	0	0	0	0	0	0	0	0	1	
0	0	0	1	1	0	0	1	1	1	1	
0	0	1	0	0	0	1	0	0	1	0	
0	0	1	1	0	0	0	0	1	1	0	

0	1	0	0	1	0	0	1	1	0	0	
0	1	0	1	0	1	0	0	1	0	0	
0	1	1	0	1	1	0	0	0	0	0	
0	1	1	1	0	0	0	1	1	1	1	
1	0	0	0	0	0	0	0	0	0	0	
1	0	0	1	0	0	0	1	1	1	0	
1	0	1	0	1	1	1	0	0	0	1	

1	0	1	1	1	1	0	0	0	1	1	
1	1	0	0	1	0	1	1	1	1	0	
1	1	0	1	0	1	1	0	0	1	0	
1	1	1	0	1	1	1	0	0	0	0	
1	1	1	1	1	1	1	1	1	1	1	